STELLA (YUXUAN) JIANG

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EDUCATION

University of California, Irvine Expected Mar. 2027
B.S. in Computer Science | **GPA: 3.88/4.0** | Dean's Honor List
Coursework: Data Structures and Algorithms, Machine Learning, Database Management, Software Engineering

WORK EXPERIENCE

Software Engineer Intern, Applied AI | Alibaba Group (global public tech company) Jul. 2026 - Sep. 2026
*CN | Tmall Campus digital services and retail ecosystem: **800+ universities and nearly 20M students**.*

- Built a **Python/FastAPI RAG service** for campus knowledge search and evaluated BM25, BGE, and RRF retrieval methods across 778 entries and 6,562 queries, selecting BGE as the production retriever with **76.40% Top-1 and 91.34% Top-3** accuracy.
- Designed coverage gates and deterministic/LLM-based **risk guards** with shadow and enforcement modes to prevent unsupported, privacy-sensitive, and safety-sensitive responses from reaching users.
- Improved campus product discovery by combining ItemCF, field-aware BM25, dense embeddings, and Weighted RRF, increasing ranking-stage **Recall@10 from 70% to 85%** in a 1,000-request offline evaluation.
- Developed structured intent and user-profile services with **batch processing pipelines**, transforming 100M raw events into **86.3M training records for 987,992 users** while isolating offline computation from latency-sensitive services.
- Delivered four community content workflows with interchangeable model providers, Pydantic contracts, fact validation, and atomic quota management, validating reliability through **371 backend/smoke tests**, 9 browser tests, and regression testing on 999 valid posts.

Software Engineer, LLM Workflow | FessorAI, UCI Capstone Apr. 2026 - Jun. 2026
Irvine, CA | Full-stack AI education platform for structured textbook generation and in-page editing.

- Built a **six-stage LLM generation workflow** with **deterministic orchestration** to transform topics into textbook content, interactive exercises, code, images, and videos while supporting OpenAI, Gemini, Claude, and Grok through unified generation interfaces.
- Designed asynchronous generation infrastructure using **Django, Celery, Redis, and PostgreSQL** to separate long-running LLM workloads from web requests, enabling dependency tracking, durable job states, retries, failure recovery, and synchronized frontend progress updates.
- Extended a **28-tool Page Assistant** with **native function calling** and conversation memory, streaming text and tool events through Django Channels, Redis, and WebSockets into editable React components for human review.
- Resolved long-form editing failures caused by a 1,000-character context cap by passing **full-section context** into tool calls, achieving a **94% revision success rate** across 100 long-section regression cases while preserving 99.8% of non-target content.

AI Engineer Intern | NextTier Dec. 2025 - Feb. 2026
Sacramento, CA | Policy-grounded receipt and expense-report audit copilot.

- Built an **AI expense-audit copilot** that analyzed receipts and expense reports, validated transaction details, and generated policy-grounded approve/review/reject recommendations, reducing median review time from **6.4 to 2.1 minutes per report**.
- Developed a **LangChain RAG pipeline** over company expense policies using document chunking, OpenAI embeddings, vector retrieval, and context-grounded generation to improve policy-aware auditing.
- Implemented structured outputs and **LLM function calling** to extract receipt fields, flag missing data, and query expense APIs for records and duplicates; improved record-backed answer **accuracy from 82% to 91%,** reduced model calls from 2.5 to 1.6 per audit and token cost by 36%.
- Designed a modular **FastAPI/Pydantic backend** separating audit orchestration and validation logic from persistent policy/receipt storage, improving scalability and maintainability; containerized the service with Docker.
- Built an **MLflow evaluation pipeline** for synthetic expense reports, evaluating **36 configurations** across chunk size, overlap, top-k, prompt, and model settings; selected a configuration that improved accuracy from **84% to 90%**, and reduced p95 latency from **5.9s to 3.6s.**

Founder & Software Engineer | FocusTrail Dec. 2025 - Present
*Irvine, CA | UCI Startup Fund-supported local-first AI task execution coach; selected for **Alibaba Tmall Campus OPC** (5/50).*

- Built a **Gemma-powered multimodal pipeline** that converts PDFs, screenshots, and notes into structured task DAGs with explicit dependencies for execution planning, progress tracking, and recovery.
- Developed a **versioned replanning runtime** supporting dependency-aware updates, task insertions, deferrals, validation, state reconciliation, deterministic fallback, and undo operations to preserve completed work during schedule changes.
- Shipped local-first React/Node.js/Express/SSE web and macOS/Windows prototypes with rule/LLM activity classification, local/cloud model routing, and **pre-inference privacy filters** that block content from sensitive apps before any model call.

Software Engineer Intern | Join-Cheer (public enterprise software company, 1M+ users) Jun. 2025 - Sep. 2025
CN | Enterprise AI-agent platform and data services for government and financial-reporting workflows.

- Migrated an enterprise AI-agent platform from **Dify to Nuwa GPT** and built **Python/Flask REST services** on Alibaba Cloud ECS/OSS to normalize heterogeneous data sources into reusable agent tools, enabling natural-language access to government and financial reporting data.
- Standardized Docker and Kubernetes CI/CD workflows across 10+ distributed services, **reducing deployment time by ~40%**, eliminating manual rollback steps; added Function Compute health checks and runtime metrics for earlier post-migration failure detection.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL
AI & ML: Generative AI, NLP, RAG, Information Retrieval, Ranking, LLM Agents, Function Calling, Structured Outputs, LLM Evaluation, MLflow, AI Guardrails
Frontend: React, WebSocket, Server-Sent Events
Backend: FastAPI, Django, Flask, Node.js/Express, Celery, REST APIs, Asynchronous Processing, Data Pipelines
Databases: PostgreSQL, Redis, SQLite, SQLAlchemy, Vector Databases
DevOps: Docker, Kubernetes, CI/CD, GitHub Actions, Alibaba Cloud, AWS, Observability